

## Advanced (Habanero)

**Q1.** Factor the trinomial  $x^2 + 5x + 6$ .

- (A)  $(x + 2)(x + 3)$
- (B)  $x(x + 5) + 6$
- (C)  $(x + 1)(x + 4)$
- (D)  $x(x + 6) + 5$
- (E)  $(x + 3)(x + 2)$

(A)  $2x^2 - 6x + x - 3$

(B)  $(2x + 1)(x - 3)$

(C)  $(x + 1)(2x - 3)$

(D)  $2x(x -$

$\frac{32}{2}) - 3x \cdot \frac{32}{2}$

$\frac{32}{2}$

(D)  $2x^2 + 3x - 8x - 3$

**Q2.** Factor the trinomial  $x^2 - 7x + 12$ .

- (A)  $x(x - 7) + 12$
- (B)  $x(x - 3) - 4x + 12$
- (C)  $x(x - 4) - 3x + 12$
- (D)  $x(x - 3) - 4x + 12$
- (E)  $x(x - 4) - 3x + 12$

**Q6.** Factor the trinomial  $x^2 - 4x - 21$ .

(A)  $x(x + 3) - 4x - 21$

(B)  $x(x - 7) + 3x - 21$

(C)  $(x + 3)(x - 7)$

(D)  $x(x + 4) + 3x - 21$

(E)  $(x + 7)(x - 3)$

**Q3.** Factor the trinomial  $x^2 + 2x - 15$ .

- (A)  $x(x + 5) - 3x - 15$
- (B)  $x(x - 5) + 3x + 15$
- (C)  $x(x + 3) - 5x - 15$
- (D)  $x(x + 5) - 3x - 15$
- (E)  $x(x + 5) + 3x - 15$

**Q7.** Factor the trinomial  $3x^2 + 2x - 5$ .

(A)  $3x^2 - 5 + 2x$

(B)  $(3x^2 + 3x) - (x + 5)$

(C)  $(x + 1)(3x - 5)$

(D)  $3x(x -$

$\frac{53}{2}) + 2x \cdot \frac{53}{2}$

$\frac{33}{2}$

(D)  $3x(x -$

$\frac{53}{2}) + 2x$

**Q4.** Factor the trinomial  $3x^2 - 2x - 5$ .

- (A)  $3x^2 - 5 - 2x$
- (B)  $3x^2 + 2x - 5x - 5$
- (C)  $(3x^2 + 3x) - (5x + 5)$
- (D)  $3x(x -$

$\frac{23}{2}) - 5$

(D)  $3x(x -$

$\frac{73}{2}) + 16$

**Q5.** Factor the trinomial  $2x^2 - 5x - 3$ .

**Q8.** Factor the trinomial  $2x^2 + x - 6$ .

(A)  $2x^2 + x - 6$

(B)  $2x^2 - 4x + 5x - 6$

(C)  $x(2x + 1) - 6$

(D)  $x(2x - 3) + 2x + 3x - 6$

(E)  $(2x - 3)(x + 2)$

### Open Ended Questions (FRQ)

**Q9.** Factor the trinomial  $x^2 + 2x - 15$ . Justify your method and check your answer.

**Q10.** Prove that  $x^2 + bx + c = (x +$

$\frac{b}{2})^2 - (\frac{b^2}{4} - c)$ .

Hence factorize the trinomial  $x^2 + 4x + 4$ .

### Chef's Tips

- Tip 1: Review factoring of quadratic expressions
- Tip 2: Make sure students can apply the zero product property and factoring by grouping
- Tip 3: Remind students to check their answers by multiplying the factors